2D Optical Modulators

Eli Joseph Rees and Alexandra-Daria Dumitriu-Iovanescu
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The aim of this review is to summarise some of the advancements brought forth by using 2D materials to develop electro-optical devices, with a particular emphasis on graphene-based plasmonic optical modulators. We start by presenting the key features of these devices, such as a large modulation depth, high switching speed, strong light-matter interaction and small device footprint. We then provide an extensive theoretical background on optical Pauli blocking and surface plasmon polaritons, covering their underlying origin, propagation behaviour and coupling to light. We proceed by looking at some elementary geometries which support these plasmonic modes, and compare them in order to find the most promising candidates both in terms of their theoretical suitability and their experimental feasibility. We then provide an overview of some of the experimental work carried out in this field in the context of graphene plasmonic and non-plasmonic modulators, as well as present the current research aimed at exploring the potential use of other 2D materials. We finish with a summary of the benefits of these proposed technologies alongside several obstacles, and discuss the prospect of their use in the mid to long term, either as stand-alone devices or integrated into existing silicon chips.

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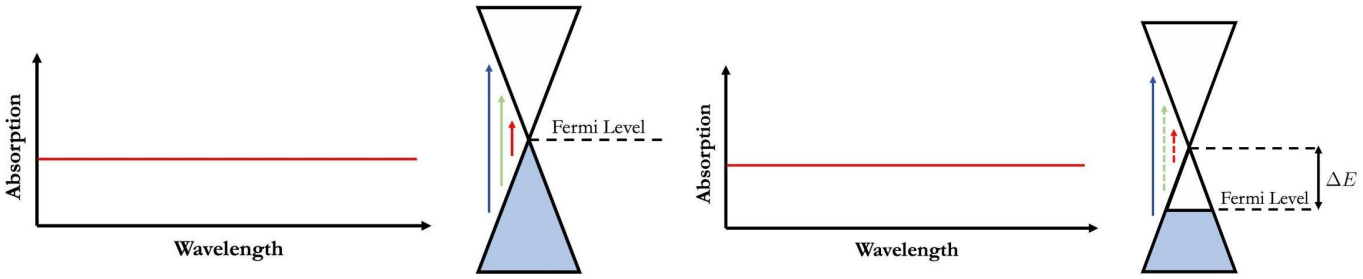
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I. INTRODUCTION

Optical modulators play a vital role in both day to day life and fundamental research. They are the basis of all fibre-optic telecommunications, where the ability to create pulses of light is key to transmitting the vast amounts of data society relies on. They are also used to tune the properties of light for myriad experiments, for example the creation of ultra-short laser pulses [1]. At their heart, optical modulators are devices that modulate (change) some fundamental property of light such as amplitude, phase, polarisation, or wavelength/frequency after the application of a control signal. These devices are already a well established and mature technology. Nonetheless, improvements are possible and the vast potential of these devices has not yet been fully realised. In this report we will explore optical modulators built using 2D crystals, how they work, their advantages, disadvantages, and how they may fit into the current technological landscape.

Optical modulators can be categorised a number of ways. One method is to categorise them by what property of light they modulate i.e. a phase modulator, an amplitude modulator, etc. Another method is to characterise devices by the type of control signal. For example, electro-optical (controlled by an electrical signal), all-optical (controlled by an additional light signal), thermo-optical (controlled by temperature changes), etc. All of these devices work according to several distinct principles and the theory behind them will be discussed later in more detail. In particular, we will focus on electro-optical and all-optical devices as the most promising candidates for telecommunications and optical computing [2].

Why might 2D crystals be interesting materials to study in relation to optical modulation? In this report we will focus on two exciting potential applications. The first is that as we shall see, 2D crystals provide a way to improve the switching speed of the modulators. To understand switching speed and its importance, consider a simple fibre-optic system where light is transmitted through the fibre at one of two amplitudes. In Binary, the lower amplitude corresponds to a 0, and the higher amplitude to a 1. It is straightforward to see that the



(a) No gating (b) Graphene gating

FIG. 1: Schematic view of the absorption spectra and band structure of graphene before and after gating. After gating, only the highest energy transition (in blue) is possible.

rate at which information can be transmitted through the wire is proportional to the rate at which the light can be changed from one amplitude to the other, this is the switching speed. Switching is often achieved by simply turning the device emitting the signal on or off. But where very long distances or high frequencies are required an optical modulator is needed [3].

The second interesting application is the use of 2D optical modulators (2DOMs) as transistors for optical computing. Optical modulators are suitable for this application as a device which changes a system or particle's state according to a control signal is effectively a transistor, and therefore a circuit can be constructed capable of performing logic.

Traditional electrical computers have enjoyed a massive improvement in computing power over many decades, with the number of transistors (and therefore computing power) increasing according to Moore's Law. However, in recent years this progress has slowed, and it is thought that mass produced computer chips are coming close to a fundamental limit in transistor density [4] [5]. As such, other technologies are needed to improve the speed of computation. One such proposed technology is optical computation, whereby the particles carrying the information are photons rather than electrons.

Such a device would have several advantages. Information could be transmitted through space faster than for traditional computers, this is an important fact as one of the major limitations on the size of high performance computing facilities is the lag between a request for data and its retrieval due to the physical distance between computing nodes. In addition, as we shall see, optical transistors are predicted to switch state faster than conventional transistors, meaning optical computers could run at higher clock speeds. One final exciting feature of optical computers is the number of degrees of freedom possessed by photons which could be utilised for information processing [6]. Analogous to how electrons carry information through voltage (and more recently, spin), photons could carry information via frequency, polarisation, phase, etc. 2DOMs are preferable over traditional optical modulators as their small size may make them easier to integrate into high density computer chips.

II. THEORY

A. Amplitude Modulation via Optical Pauli Blocking

In this section we will examine the physics and theory behind several promising phenomena that have been utilised to create 2DOMs, with a particular focus on optical Pauli blocking.

Graphene's band structure has several unusual properties that make it an exciting 2D crystal for optical modulators. Graphene's dispersion relation is linear, and in addition there is zero band gap [7]. Because of this, the absorption coefficient of graphene is given by $\pi\alpha$, where α is the fine structure constant and is equal to $\frac{e^2}{\hbar c}$ [8]. Aside from being defined by fundamental physical constants, what is notable about this fact is that the graphene absorbs 2.3% incident light over a wide range of wavelengths, including the regions used for telecommunications.

This constant absorption spectra can be understood by examining what happens when a photon with a particular energy E is incident on a graphene sheet. Since graphene's band structure has no band gap there will always be two states of equal k in the valence and conduction bands, respectively, with a difference in energy equal to E, and thus an electron will be present capable of absorbing the photon and becoming excited.

Consider now what would happen were a voltage applied, the Fermi energy would be raised or lowered by a small amount ΔE, as shown in Figure 1. If a low energy photon is incident on the graphene sheet it will not be possible to excite an electron, as either the electron's initial state will now be unoccupied, or its final state will already be occupied. This will be true for all photons

with energies less than $2\Delta E$. This is perhaps illustrated more clearly in Figure 1 where in the top diagram all three transitions are allowed, but in the bottom diagram the red and green transitions are no longer possible.

Therefore, by applying a gating voltage to a graphene sheet we can alter the absorption characteristics of the device. The dependence of the absorption on the wavelength is shown in Figure 1(b). Due to the low number of charge carriers in undoped graphene, large changes in the Fermi level can be induced by relatively small voltages [9], but this effect can also be achieved without applying a voltage. Instead, another light source is shone at the graphene to excite a population of electrons. A subsequent light pulse with the same energy will not be absorbed, as the electrons that would interact with the light are already excited. Both phenomena are known as optical Pauli blocking and are the basis for many 2D electro-optical and all-optical modulators.

The change in a signal strength due to an optical modulator is known as modulation depth and is often quoted in either percent or decibels. Whilst 2.3% absorption is significant for a single layer of atoms, most applications such as telecommunications require a significantly higher modulation depth of at least 50% [10]. If the modulation depth is too low, it is hard to distinguish between the signal and noise. Many devices have been designed with the aim of improving the light-matter interaction strength of the system and therefore increasing the modulation depth. These approaches range from phase changing interferometers, to waveguides, to cavity enhancement [11] [12]. One of the most exciting approaches which we will now discuss is the use of plasmons.

B. Plasmonic Modulators

Since electromagnetic waves induce electric fields perpendicular to their direction of propagation, in-plane incident light requires electrons to oscillate out of the plane [13]. However, due to the 2D nature of the material this is not possible, as electrons are confined in the plane of the graphene sheet, and without being able to oscillate, they will not contribute to absorption. The direction of the propagating light is shown in Figure 2 by the wiggly oriented line, and the direction of the electric field and therefore the electron oscillation in blue.

Looking at this figure we can see that in order to get the maximum absorption of the incident light, this should fall transversely onto the layer. However, such a type of modulator cannot unfortunately fully utilise graphene's potential, because of the very small interaction surface affected by graphene.

Thus a way to have both a large interaction area and a simultaneously strong in-plane electric field is to use the

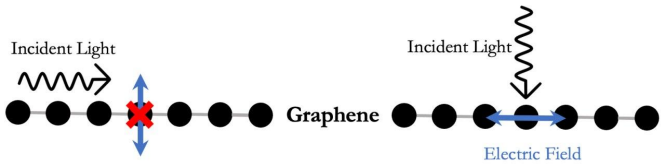


FIG. 2: Graphene sheet with incident light at different orientations: parallel to it, but inducing out-of-plane electron oscillations which are not allowed due to the 2D confinement (on the left), and perpendicular to it, affecting only a small interaction area (on the right).

incident light to excite certain modes which naturally propagate along the surface.

In 1956, David Pines theoretically described the energy losses experienced by fast electrons travelling through metals as being due to collective oscillations of free electrons in the metal, which he called 'plasmons' [14]. In the same year, the term 'polariton' for used for the first time by Ugo Fano to describe the coupled oscillation of bound electrons and light inside transparent media [15]. Later it was found that plasmonic modes could also exist near the surface of metals, which led to the discovery of the surface plasmon polariton (SPP) by Steve Cunningham [16].

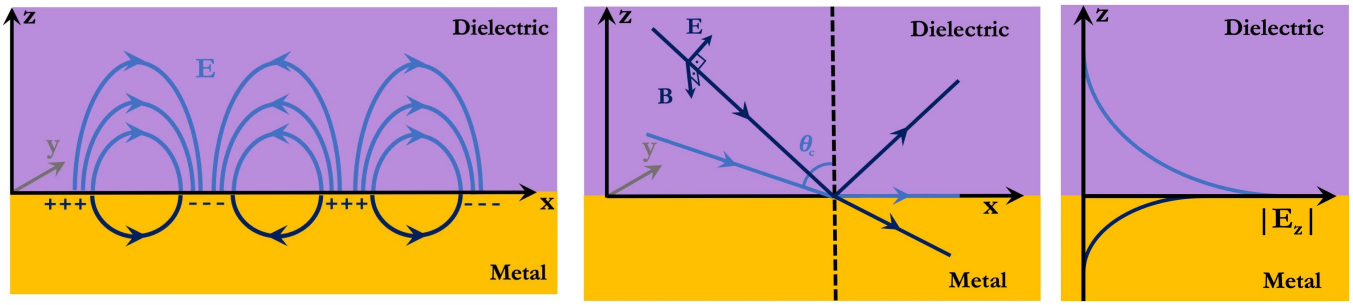
SPPs are electromagnetic excitations which propagate in a wave-like fashion along the planar interface between a metal and a dielectric medium and whose electromagnetic field is confined to the near vicinity of the interface [17]. To better explain how these are used for electro-optical devices, we start by discussing some fundamental concepts of SPPs, such as their excitation, coupling and propagation behaviour.

Over the past decade, plasmonic nanoscale devices have been investigated and revealed to provide both high energy efficiency due to the enhanced light-matter interaction and small device footprints, beneficial for dense device integration [18].

C. Surface Plasmon Polaritons

SPPs are surface electromagnetic waves that propagate along the interface between a metal and a dielectric material, as shown in Figure 3(a). If we consider light refracting at the interface of these two materials with refractive indices n_m and n_d, Snell's law allow us to calculate the critical angle at which refraction can still occur. A simple sketch of this is shown in Figure 3(b).

Although a wave at an incident angle larger than the critical one is totally reflected at the interface, this wave has more momentum along the surface plane than can be supported by the metal. This results in oscillating



(a) Surface plasmon polariton (b) Reflection and refraction (c) Evanescent field

FIG. 3: SPP excited at the interface between a metallic and a dielectric media (on the left), reflection and refraction of an electromagentic wave at the same interface, where θ_c is the critical angle (in the middle) and the evanescent electric field in the z-direction (on the right). Images adapted from Li *et al.* [19] and from Zhang *et al.* [20].

charges at the interface with associated radiation fields penetrating into the metal, fields which are spatially decaying (evanescent) in the direction normal to the interface, as shown in Figure 3(c). These evanescent fields for the incident wave beyond the critical angle are useful for coupling radiation to SPPs [20].

To fully characterise SPPs, we need to determine their dispersion relation. In order to do this, we need to identify the fields at the interface and with the help of Maxwell's equations to calculate the momentum of the propagating plasmonic mode.

The boundary conditions for electromagnetic fields state that there is no net surface charge or current on the interface, then the tangential component of the electric field $\boldsymbol{E}$ and that of the magnetic field strength $\boldsymbol{H}$ is continuous across the interface [13]:

$$E_{x/y}^{(d)} = E_{x/y}^{(m)} \tag{1}$$

$$H_{x/y}^{(d)} = \frac{1}{\mu_d \mu_0} B_{x/y}^{(d)} = \frac{1}{\mu_m \mu_0} B_{x/y}^{(m)} = H_{x/y}^{(m)} \tag{2}$$

together with the normal component of the magnetic field $\boldsymbol{B}$ and that of the displacement field $\boldsymbol{D}$ [13]:

$$B_z^{(d)} = B_z^{(m)} \tag{3}$$

$$D_z^{(d)} = \varepsilon_d \varepsilon_0 E_z^{(d)} = \varepsilon_m \varepsilon_0 E_z^{(m)} = D_z^{(m)} \tag{4}$$

where d and m are used to label the dielectric and metallic media, respectively.

The electric and magnetic fields can therefore be written in terms of their components as:

$$\boldsymbol{E} = \begin{cases} \boldsymbol{E}^{(d)} = \left(E_x^{(d)}, 0, E_z^{(d)}\right) e^{-k_z^{(d)} \cdot z} \, e^{i(k_x x - \omega t)}, & \text{when } z > 0 \\ \boldsymbol{E}^{(m)} = \left(E_x^{(m)}, 0, E_z^{(m)}\right) e^{k_z^{(m)} \cdot z} \, e^{i(k_x x - \omega t)}, & \text{when } z < 0 \end{cases} \tag{5}$$

$$\boldsymbol{B} = \begin{cases} \boldsymbol{B}^{(d)} = \left(0, B_y^{(d)}, 0\right) e^{-k_z^{(d)} \cdot z} \, e^{i(k_x x - \omega t)}, & \text{when } z > 0 \\ \boldsymbol{B}^{(m)} = \left(0, B_y^{(m)}, 0\right) e^{k_z^{(m)} \cdot z} \, e^{i(k_x x - \omega t)}, & \text{when } z < 0 \end{cases} \tag{6}$$

where $k_{x/y}$ is the momentum in the x or y direction, and ω is the frequency of the propagating wave [21]. It is important to note that since we have total internal reflection at the interface as shown in Figure 3(b), both the momentum component along the x-direction k_x and the frequency of the incident wave ω are conserved, and therefore do not require further labelling m or d.

After solving Gauss's law $\boldsymbol{\nabla} \cdot \boldsymbol{E} = 0$, the z-components of the electric field take the form

$$E_z^{(d)} = i \frac{k_x}{k_z^{(d)}} E_x^{(d)} \quad \text{and} \quad E_z^{(m)} = -i \frac{k_x}{k_z^{(m)}} E_x^{(m)}. \tag{7}$$

In order to get the dispersion relation of the SPPs, we also need to make use of Faraday's law of induction $\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$, where the magnetic field at the interface is given in Equation (6). This yields two relations:

$$E_x^{(d)} \left(k_x^2 - k_z^{(d)2}\right) = i k_z^{(d)} \omega B_y^{(d)} \tag{8}$$

$$E_x^{(m)} \left(k_x^2 - k_z^{(m)2}\right) = -i k_z^{(m)} \omega B_y^{(m)}. \tag{9}$$

The electric field given in Equation (5) must obey the electromagnetic wave equation $(v^2 \nabla^2 - \partial_t^2)\boldsymbol{E} = 0$, where v is the phase velocity and it depends on the relative

permittivity and permeability of each media via relations $v_d = c/\sqrt{\mu_d \varepsilon_d}$ and $v_m = c/\sqrt{\mu_m \varepsilon_m}$. Solving the wave equation imposes the conditions

$$k_x^2 - k_z^{(d)2} = \mu_d \varepsilon_d \, (\omega/c)^2 \tag{10}$$

$$k_x^2 - k_z^{(m)2} = \mu_m \varepsilon_m \, (\omega/c)^2 \tag{11}$$

when $z > 0$ and $z < 0$, respectively.

Taking the ratio of Equations (8) and (9) and using the boundary conditions at the interface $E_x^{(d)} = E_x^{(m)}$ and $\mu_m B_y^{(d)} = \mu_d B_y^{(m)}$, we obtain:

$$\frac{\mu_m}{\mu_d} \cdot \frac{k_x^2 - k_z^{(d)2}}{k_x^2 - k_z^{(m)2}} = \frac{\varepsilon_d}{\varepsilon_m} = -\frac{k_z^{(d)}}{k_z^{(m)}} \tag{12}$$

which by plugging into the two relations given in (10) and (11) gives us the relation for the momentum k_x.

Since these equations describe the SPP which is propagating along the surface in the x-direction, the momentum of our surface plasmon polariton must be eaxctly k_x. By rearranging Equation (12) for k_x and considering only non-magnetic media, i.e. $\mu_d = \mu_m = 1$, we get

$$k_{spp} = k_x = \frac{\omega}{c} \sqrt{\frac{\varepsilon_d \, \varepsilon_m}{\varepsilon_d + \varepsilon_m}}. \tag{13}$$

We reach our final expression by using the Drude model approximation for the frequency-dependent dielectric function of a metal $\varepsilon_m(\omega) = 1 - (\omega_p/\omega)^2$, where ω_p is the bulk plasma frequency [22]. This enables us to rewrite Equation (13) as:

$$k_{spp} = \frac{\omega}{c} \sqrt{\frac{\varepsilon_d \, (\omega^2 - \omega_p^2)}{(1 + \varepsilon_d) \, \omega^2 - \omega_p^2}}. \tag{14}$$

This dispersion relation is illustrated in Figure 4, which better shows the nonlinear characteristic of this plasmonic mode. Because of this feature, for every given frequency the momentum of the SPP wave is always larger than that of the light. This momentum mismatch needs to be overcome in order to couple the light to the SPP mode at the interface, i.e. to excite the plasmonic mode of the desired frequency [17].

D. Coupling Light to SPPs

There are various ways in which the momentum of light and that of the SPP can be matched, by using either photon tunneling during total internal reflection or diffraction effects [17].

In the configuration shown in Figure 5(a), the light undergoes total internal reflection at the interface between the metallic film (shown in yellow) and the prism

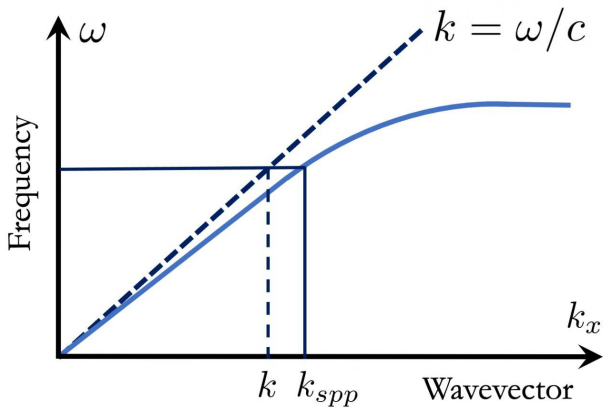


FIG. 4: Dispersion relation of SPPs, shown in blue. For a given frequency ω, the plasmonic mode carries wavevector k_{spp}, while light has wavevector $k < k_{spp}$.

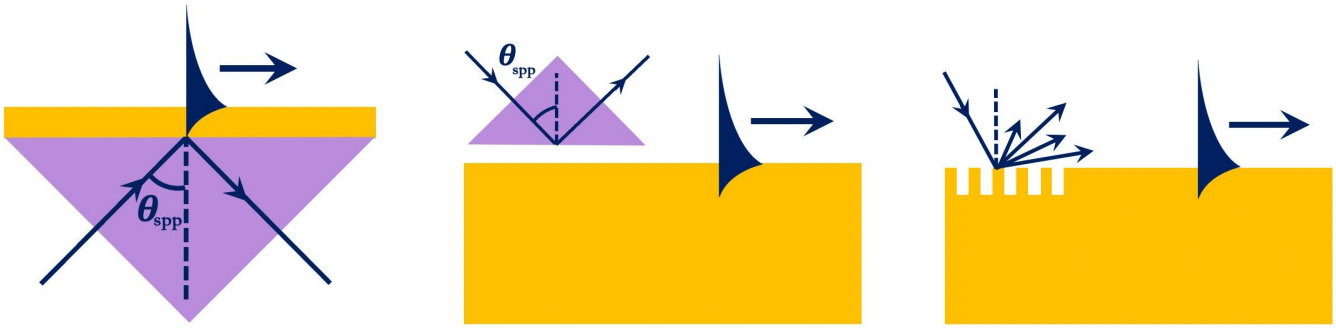
made of a dielectric material (shown in purple). At a specific angle of incidence θ_{spp}, the in-plane component of the light wavevector inside the prism is equal to the SPP wavevector at the air–metal interface, such that the light of this corresponding resonant frequency can tunnel through the metal film (thanks to the evanescent field) and be coupled to the surface plasmon polaritons [20]. This is called a Kretschmann configuration and was first proposed in 1968 [23][24].

In the case of a thick metal film or the surface of a bulk metal, this configuration cannot be used, as the evanescent SPP field is decaying away from the interface. However, if the prism is placed close to the metal surface, but without touching it, then the photons can tunnel through the air gap between the prism and the surface. This is called an Otto configuration [25] and is sketched in Figure 5(b).

Figure 5(c) shows another way to provide the wavevector conservation for SPP excitation, by using diffraction effects. This diffracting grating separates the incident light into different wavelengths, such that the components of the diffracted light whose wavevectors coincide with the SPP wavevector are coupled to the surface plasmon polaritons. Although used prior to them, Tetin *et at.* numerically investigated this light-SPP coupling process and found that an efficiency of $\approx 24\%$ could be obtained at telecom wavelengths via diffraction grating [26].

III. GEOMETRY

The three configurations shown in Figure 6 can be considered elementary building blocks for any plasmonic waveguide, therefore analysing each of them in depth will both reveal the main physics involved in the process and also hint on the most promising geometry to consider for graphene-based modulators [27].



(a) Kretschmann configuration (b) Otto configuration (c) Diffraction grating

FIG. 5: Light-SPP coupling: placing a prism made of a dielectric material below a metallic thin film (on the left) or next to the metal, while leaving a small air gap (in the middle), or using a diffraction grating (on the right). Image adapted from Zhang *et al.* [20].

A. Modulation Depth

As already mentioned, one of the figures of merit we are most interested in is the modulation depth that the device can achieve.

When trying to compare different geometries, it is beneficial to make some crude estimations which are aimed at simplifying the calculation without changing the general description of the system. In our case, we might neglect the dielectric layer between graphene and the metal (shown in purple in Figure 6), by assuming the two are in contact. We also consider that the gating obtained using optical Pauli blocking changes the state of graphene from maximum absorbing to completely non-absorbing.

The modulation depth per unit length in logarithmic dB units is given by [27]:

$$M = \frac{-10\log_{10}(1-\Delta W/W)}{\Delta x} \approx \frac{10}{\ln(10)}\frac{\Delta W}{W\Delta x} \tag{15}$$

where W is the energy flow through a cross-sectional area perpendicular to the mode propagation direction Δx, and ΔW is the additional absorption of energy introduced by the graphene layer.

During a period of time Δt and along the propagation direction Δx, the additional absorption of energy induced by the graphene layer is

$$\Delta W = \underbrace{\mathrm{Re}(\sigma)}_{c\alpha/4}\Delta x\,\Delta t\int E_{\parallel}^2\,\mathrm{d}\,l \tag{16}$$

where $E_{\parallel}$ is the tangential electric field on the metal surface and therefore in-plane with the graphene layer. The real part of the conductivity of graphene is written explicitly and α is its fine structure constant.

When the tangential electric field $E_{\parallel}$ is roughly constant along the direction of integration, we can simplify Equation (16) as $\int E_{\parallel}^2\,\mathrm{d}\,l \approx E_{\parallel}^2\Delta l$, where Δl is either the plasmonic mode width or the length of graphene layer in the direction perpendicular to the mode propagation, depending on which one is narrower. This suggests that in order to obtain a good modulation depth, we would like to have a large overlap length between the graphene layer and the plasmonic mode field.

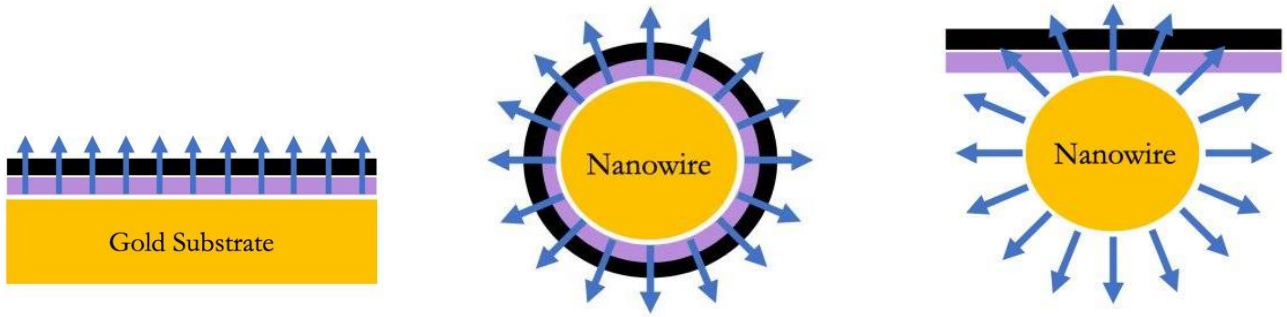
From Equations (15) and (16) it becomes clear that the modulation depth per unit length scales as the strength of the electric field squared, but only the electric fields in plane with the graphene sheet contribute to the absorption. This is not surprising as we already discussed the mechanism behind this back in Section II B. Since we have two variables which we want to maximise simultaneously, i.e. the extent of in-plane electric field and its strength, we are going to look at some of the most common geometries for plasmonic modulators and compare the modulation depths obtained in each of them.

B. Planar SPPs

Figure 6(a) shows a graphene sheet sitting on top of a flat metal surface, which is the simplest configuration which supports planar surface plasmon polaritons.

The modulation depth in this case was calculated by Radko *et al.*, who showed that it scales as the ratio of the tangential and longitudinal electric fields, squared. Assuming a choice of metal for which $\varepsilon_m \sim 100$ at telecom wavelengths $\lambda = 1.5$ µm, they approximated the modulation depth as $M \approx 1.7\times10^{-3}$dB/µm [27]. The order of magnitude of this result is expected, since the chosen geometry favours transverse electric fields which are not contributing to the modulation.

This is an optimistic start since it tells us that we can



(a) Planar graphene sheet (b) Nanowire wrapped in graphene (c) Nanowire adjacent to flat graphene

FIG. 6: Three geometries which support either planar or cylindrical SPPs: graphene sheet sitting on top of a flat metal surface (on the left), graphene sheet wrapping a metal nanowire (in the middle) and planar graphene sheet touching metal nanowire (on the right). The layer of graphene in shown in black, the dielectric used for gating in purple, the metal in yellow and the electric field lines in blue. Image adapted from Radko *et al.* [27].

indeed achieve modulation by optical Pauli blocking in graphene at the desired telecom wavelenghts. However, this modulation alone is too small to be used in applications, so we should investigate other geometries as well.

C. Cylindrical SPPs, full graphene wrapping of metallic nanowire

The second configuration, shown in Figure 6(b), shows graphene wrapping a nanowire, with the electric field lines poinitng outwards from the nanowire. Although at first glance this might not seem too different from the previous case, it is important to note that the plasmonic mode of cylindrical SPPs is more tightly confined than for planar ones.

By introducing a nanowire instead of a flat substrate, we now have an additional degree of freedom to exploit, namely its radius. In this case, the electric field penetration depth in the nanowire is directly proportional to the wavelength of the plasmonic mode, where we denote this dimensionless proportionality constant ξ.

After approximating the additional absorption introduced by graphene in this geometry, Radko *et al.* obtained a value of $M \approx 6.7 \times 10^{-3}\xi^{-3}$ dB/µm. For thin nanowires of 20 nm radius they claim that $\xi \sim 0.15$, while for thicker nanowires of radius 150 nm, $\xi \sim 0.55$.

This expression yields modulation depths between $M \approx 0.04$ dB/µm and $M \approx 2$ dB/µm, depending on the thickness of the nanowire [27]. This result shows a great improvement compared to the previous section, but wrapping graphene around such small are fragile nanowires might not be feasible in experiments.

D. Cylindrical SPPs, flat continuous graphene sheet touching metallic nanowire

The last configuration among these three is sketched in Figure 6(c) and it shows another metal nanowire, but this time only touching a planar graphene sheet. The main difference now compared to the previous cases is that the tangential electric field can no longer be considered constant, but it should be carefully integrated along the graphene layer.

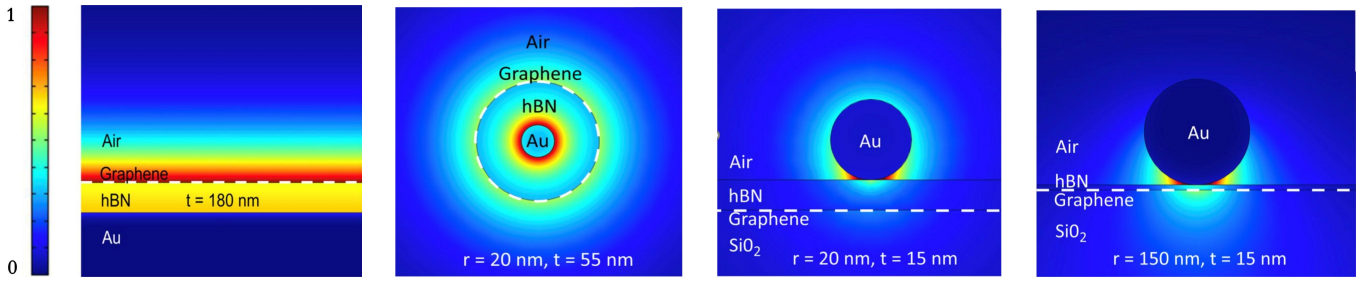
In order to do this, two approximations must be made: first, the penetration depth of the plasmonic mode must be at least the same length as the nanowire diameter, and second, the SPP mode is fully wrapped around the nanowire: $\Delta l \approx 2\pi r$.

In a similar fashion to the earlier section, we consider both thick $r \sim 150$ nm and thin $r \sim 20$ nm nanowires, configurations for which Radko *et al.* calculated modulation depths of $M \approx 0.4$ dB/µm and $M \approx 3.2$ dB/µm, respectively [27]. This is for sure the most promising among the three geometries, but it is also worth analysing how our initial simplifications impact these results and whether more realistic estimations can be calculated.

E. Numerical Modelling

As stated before, at the core of these devices is graphene's potential to act as a modulator for certain wavelengths via optical Pauli blocking. However, in order to increase or decrease the Fermi level in graphene for blocking specific transitions from happening, graphene needs to be gated with a dielectric material, which is aimed at positively or negatively doping it.

The major simplification we did so far was to ignore



(a) Planar SPP (b) Cylindrical SPP (c) Flat sheet of graphene, thin nanowire (d) Flat sheet of graphene, thick nanowire

FIG. 7: Temperature colour plots of the strength of the electric field for the Au/hBN/graphene/air geometrical configurations considered. Images are taken from Radko *et al.* [27].

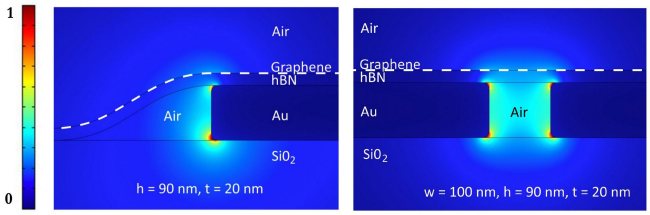
the presence of this dielectric spacer and to assume that the metal was in direct contact with graphene. However, the presence of this separator moves graphene away from the metal-dielectric interface where the electric fields are enhanced, which can strongly reduce the achievable modulation depth [27]. At the same time, including this layer allows us to freely select its thickness, which is another degree of freedom that we are able to exploit in order to maximise the light absorption.

The choice of the dielectric material is crucial before modelling how the strength of the electric fields changes the modulation depth. Since hBN has a large resistivity, very good elastic properties, high breakdown field strength, while also being feasible to use for the fabrication of atomically-flat layers [28][29], it is the desired candidate for graphene gating.

As a continuation of the analysis made in the previous sections, Radko *et al.* also used a very precise finite-element method to model the electric fields and to simulate the modulation depths of these configurations when including the hBN layer. The temperature colour maps of the electric field strength that they produced are shown in Figure 7. They show the Au/hBN/graphene/air layers for the planar SPPs - Figure 7(a), cylindrical SPPs when graphene is wrapping the nanowire - Figure 7(b), and also cylindrical SPPs for both thin and thick nanowires next to a planar graphene sheet - Figures 7(c) and 7(d).

For the planar geometry, they found that a thickness of the hBN layer of around $t = 150$ nm would correspond to the maximum modulation depth. Intuitively, the fact that there is an optimal nonzero thickness can be understood by thinking of the energy of the plasmonic mode stored in the dielectric layer: for a very thick layer, the evanescent field reaching graphene from the metal-dielectric interface would be very small, while a very thin layer cannot store the energy of the plasmonic mode, so the electric field is pushed into the air.

For a nanowire fully wrapped by graphene, they also



(a) Wedge plasmonic mode (b) Slot plasmonic mode

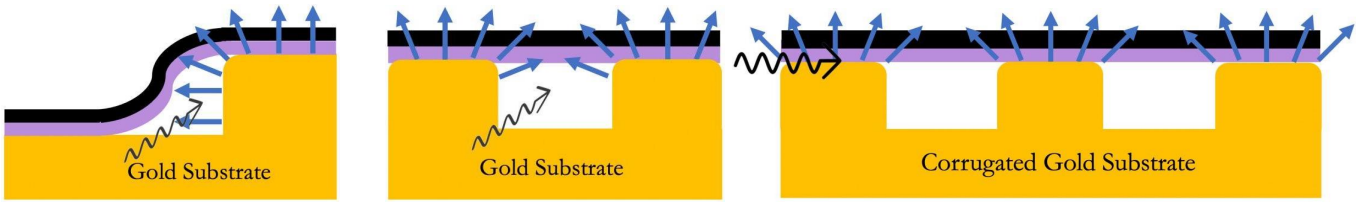
FIG. 8: Plots of the strength of the electric field for the two Au/hBN/graphene/air edge-based configurations. Images are taken from Radko *et al.* [27].

found an optimal thickness $t = 55$ nm, but the simulation suggested this geometry provides an unfortunate trade-off between the plasmonic mode confinement and the extent of in-plane electric field [27]. This is mainly due to the fact that this configuration forces graphene to be exactly perpendicular to the strongest electric field, so that it cannot fully exploit the potential of this mode.

This issue is partially resolved in the third geometry, when most of the transverse electric field lines have a nonzero projection on the graphene plane. In this case, however, the optimal thickness of the dielectric turns out to be zero. Although this scenario is impossible, as explained previously, good modulation depths were still predicted for a thickness of around $t = 15$ nm [27], which is experimentally feasible.

Finding the perfect geometry is not enough if it is not also experimentally feasible or if it imposes particularly challenging restrains on the fabrication process. Despite the substantial research in the field of nanowire photonics [30][31][32], manipulating and positioning single nanostructures still remains a major obstacle.

Luckily, edge-based waveguides manage to approximately replicate the geometry of the nanowire adjacent to the graphene sheet discussed in Section III D, while providing more flexibility in fabrication and consequently



(a) Wedge plasmonic mode (b) Slot plasmonic mode (c) Corrugated plasmonic mode

FIG. 9: Three geometries which support the plasmonic modes considered in this section, where the directions of the mode propagation are given by the wiggly oriented lines. The layer of graphene in shown in black, the dielectric used for gating in purple, the metal in yellow and the electric field lines in blue. Image adapted from Radko *et al.* [27].

easier integration with other nano-optic components [27].

F. Edge-based Configurations

In 2005, Pile *et al.* theoretically predicted and experimentally observed strongly localised plasmons around a metal wedge [33], which they already expected to be good candidates for subwavelength plasmonic waveguides. Only three years later, two different groups led by Moreno and by Boltasseva continued the research on the theoretical and on the experimental front, respectively, and reported wedge plasmonic waveguides functioning at telecom frequencies [34] [35].

A clear way to understand the similarities between edge-based waveguides and the previous configuration with the flat graphene layer touching the nanowire is to look at the electric field strength plot of this new geometry. Radko *et al.* performed the same numerical modelling on modulators based on edge plasmonic modes, propagating either along or transverse to the edge. Figure 8(a) shows the electric field of a spatially limited planar gold film deposited on a dielectric substrate, with a graphene flake covering the edge. Figure 8(b) also shows a similar configuration, but this time with gold deposited on each side of the air gap and with graphene sitting on top of the newly formed slot.

Comparing these maps with what we expected from the electric field on the flat layer graphene due to the gold nanowire - Figures 7(c) and 7(d) - we can see they indeed approximate very well the previous geometry. In addition to avoiding the manipulation of individual nanowires altogether, the main advantage of edge-based devices is that these specific shapes can be experimentally realised using lithography, which could potentially also make them feasible for large-scale fabrication.

Three possible configurations which support these edge plasmonic modes are shown in Figure 9, where the main difference between them is the direction of propagation of the mode, which is represented by the wiggly oriented lines. While in the first two geometries 9(a) and 9(b) the SPP propagates along the corner edge of the gold substrate or through a slot composed of two of these edges, in the third one shown in 9(c), the SPP propagates perpendicular to the corrugated features.

IV. EXPERIMENTAL WORK

A. Device Characterisation

When assessing the quality of any optical modulator several quantitative and qualitative factors must be considered. Firstly, the switching speed is the frequency at which modulation can be applied and is of vital importance for most applications. Consider a sheet of graphene on top of a metal substrate with an insulating layer of hBN inbetween. This is effectively a capacitor with an atomically thin dielectric. The characteristic time constant for such a capacitor is given by $\tau = RC$, where R is the resistance and C the capacitance. Because the inter-plate spacing is so small the capacitance is massive, giving a switching speed of approximately 500 GHz as a theoretical limit.

Secondly, the modulation depth (sometimes quoted as modulation depth per unit length) must be sufficiently high for signal processing - an acceptable but not desirable modulation depth of 50% ($\approx$ 3 dB) may be taken as a benchmark. Related to modulation depth is the shape of the modulation over time. Ideally, the modulation of the device would change like a step function, with sharp transitions that clearly delineate between states. In reality, however, such transitions are not instantaneous but instead spread out in time, how sharp these transitions are may be more or less relevant depending on the application.

Thirdly, the range of wavelengths over which the device functions must be tailored to the application. Telecommunications typically operate in the terahertz region and a wide bandwidth is desirable. Whereas for a modulator being used to generate ultra-short laser pulses a wide bandwidth may not be required.

Finally, qualitative factors such as manufacturing difficulty, robustness, and device footprint must be considered. If 2DOMs are to ever become widespread and commercially viable it must be feasible to manufacture them at scale, and they must function sufficiently after a long period of time and over many duty cycles. Whilst this information is typically not reported it is vital when comparing devices and promising avenues of exploration.

B. Non-plasmonic Graphene Devices

Silicon waveguides are a mature technology and widely used for the manipulation of light at the nanoscale. Many devices have been designed utilising graphene on a silicon substrate [11][36][37]. Although not a plasmonic device, one of the first 2DOMs using a silicon substrate was reported by Liu *et al.* in 2011 [38]. The device consisted of a graphene sheet draped over a silicon waveguide coated in 7 nm thick Al_2O_3 layer. By changing the voltage between the waveguide and the graphene sheet modulation was achieved via optical Pauli blocking. The device had a switching speed of 1GHz, and an operating range of between 1.35 to 1.6 µm. Whilst the device was relatively small for the time at 25 μm^2, the modulation depth was poor at just 0.1 $dB\,\mu m^{-1}$. This is $\approx$ 2.3% μm^{-1} and is therefore comparable to a simple graphene sheet without additional architecture. The modulation change was not particularly sharp, which Liu *et al.* attribute to defects in the graphene sheet. It should be noted the graphene sheet was grown using CVD which is known to contain a higher density of impurities compared to mechanically exfoliated graphene, which may have produced sharper modulation transitions [39].

There are also graphene based devices designed to integrate with existing fibre-optic systems. Li *et al.* produced an all-optical modulator by heating a fibre-optic cable and pulling it to produce a very thin cable 10 µm in diameter, this cable was then clad in graphene [40]. using this method the authors were able to achieve 38% modulation depth using 20 µm of graphene cladding. Plasmonic devices require an antennae to convert incident light into plasmons and vice versa, this step leads to losses in signal intensity. A modulator such as Li *et al*'s - one wrapped around a fibre optic cable - bypasses the losses associated with plasmon excitation, and would have the advantage of being able to fit into existing infrastructure more easily than plasmonic devices. This comes at the cost of weaker light-matter interactions however, and would therefore necessitate a larger device footprint.

C. Graphene Plasmonic Devices

Plasmons have been both theoretically and experimentally to increase the light-matter coupling in graphene and 2D crystals in general [41][42][43]. To investigate which type of Plasmon might be most appropriate for a 2DOM Ansell *et al.* created a chip which guided SPPs along waveguides with different geometries to enhance the in plane electric field strength along the graphene sheet [44]. As shown in Figure 10, the chip consisted of a Gold substrate with three geometries etched into it and a graphene sheet insulated by a layer of hBN on top of it. The three geometries on the substrate correspond to three distinct plasmonic modes, each of which was characterised. The three plasmonic modes analysed were free plasmons which were formed in the flat region of the chip, corrugated plasmons which were formed in the corrugated part of the chip, and wedge plasmons which formed at the edge of the chip. All these geometries were analysed in detail in the previous sections.

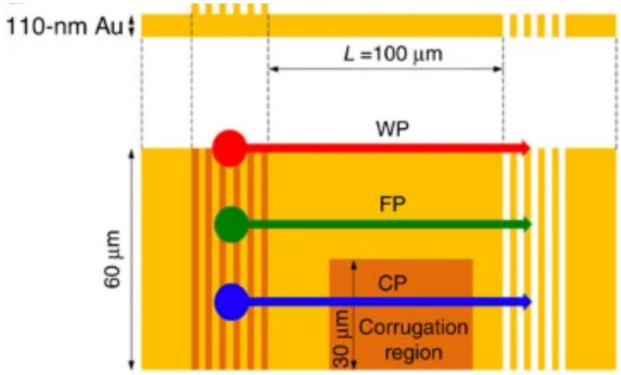


FIG. 10: Ansell *et al.*'s device testing the interaction strengths of three modes of plasmons: corrugated (**blue**), free (**green**), and wedge (**red**) [44].

The device was designed for operation in the terahertz fibre-optic region and had a significantly smaller footprint compared to the silicon waveguide of just 10 μm^2. As predicted by Radko *et al.*, of the three plasmonic modes the wedge plasmons displayed the best modulation depth due to the enhanced in plane electric fields [27]. However, this modulation depth was an order of magnitude lower than that of the silicon waveguide at just 0.03 dB μm^{-1}. The authors note however that because of the geometry of the sample it was impossible to excite just the wedge plasmonic mode, and that therefore the modulation depth quoted includes a significant contribution from the excitation of the less effective free plasmonic modes. Whilst these results display poorer modulation depth than non-plasmonic waveguides, silicon waveguides are significantly more mature and the device investigated was not dedicated to the excitation of wedge plasmons. As such, a device specifically designed to excite wedge plasmons is predicted to perform significantly better.

In 2018 Ono *et al.* produced an all-optical device which utilised an additional pulse of light to control the

distribution of electrons in the graphene sheet [45]. In all-optical modulators the switching speed is generally limited by the relaxation time of the excited electrons. In graphene, however, the relaxation time is very short [46], and this allowed for the generation of a pulse with a width of just 260 fs, which if run continuously would correspond to a switching speed of $\approx$ 4 GHz. However, continuous pulse cycles were not carried out which limits the relevance of this modulator so far as telecommunications are concerned. The device utilised an extremely narrow waveguide ($\approx$ 30 nm) to create subwavelength plasmonic modes of the kind described by Veronis *et al.* and Pile *et al.* [47] [48]. Using these plasmonic modes to couple with the graphene sheet above the waveguide the device produced a modulation depth of 3.5 dB over just a 4 µm interaction length (0.875 dB μm^{-1}), a very strong modulation. Whilst the modulation depth is impressive and the authors display confidence in the ability of the device to run continuously for many consecutive pulses the work has not been done as of yet.

One of the disadvantages of all-optical modulators is that additional energy is required to produce the control signal light needed to excite electrons in the graphene, this is important when considering the potential power consumption of all-optical computing chips. However, the narrow nature of the waveguide reduced the number of electrons that must be excited, with the authors reporting a switching energy of 35 fJ, four orders of magnitude smaller than for similar all-optical graphene devices.

D. Other 2D Materials

Although historically a lot of the research has focused on the use of graphene, there has been much recent work examining the potential of other 2D crystals, such as many of the transition metal dichalcogenides (TMDs) [49], and black phosphorous [50] [51].

TMDs do not have the same linear dispersion relation or zero band gap properties as graphene and instead posses direct band gaps between 1 - 2.5 eV [52]. TMDs also exhibit strong excitonic properties which manifest as large absorption peaks across a range of wavelengths. These excitonic properties have been utilised to create modulators with very strong light-matter couplings. TMDs also display novel saturable absorption properties which allow them to be used to create ultra-fast laser pulses [53][54].

Klein *et al.* produced a device which utilised the strong interaction between SPPs and excitons found in the WSe_2 to produce modulation depth of 73% along a 4 µm WSe_2 layer [55]. Such a strong modulation suggests research into TMD-2DOMs is also a very promising direction of study.

V. CONCLUSIONS AND OUTLOOK

Graphene has been shown to be remarkably useful for the construction of 2D optical modulators. Its unique band structure allows direct modulation of its absorption characteristics via optical Pauli blocking. Graphene displays very large absorption for a single layer of atoms, and it has been reported that its inherent thinness lends itself well to modulators with exceptionally small footprints. This thinness also leads to very high theoretical switching speeds, though to date device switching speeds have not exceeded those of traditional optical modulators.

Despite these promising characteristics, several key challenges remain. The devices are difficult to make and tend to be fragile. Whilst the technology is still in its infancy, the robustness of these devices over many duty cycles is a key consideration for future applications. In addition, although graphene's absorption coefficient is excellent for a single layer of atoms, its unenhanced modulation depth is not sufficient for use in telecommunications or optical computing.

As such, plasmons have been utilised to enhance the light-matter interaction between the signal and graphene to great success. There is an extremely large parameter space to explore in relation to how the in-plane electric field component of plasmons can be improved, such as surface composition, surface geometry, dielectric material, and methods of plasmon excitation. This large parameter space is exciting as the progress of plasmonic devices with respect to modulation depth has been dramatic over the last few years, going from barely better than unenhanced graphene to well above the 50% modulation depth benchmark required for many applications. The variety of potential plasmonic modes available suggests device performance will improve rapidly in the coming years. In addition to that, other 2D crystals such as TMDs are a very promising avenue of research owing to the strong interaction between excitons and plasmons.

Although it is not expected that all-optical chips will be produced in the near future, there is a possibility that modulators based on 2D crystals could be integrated into existing silicon chips as a first step. Examples of how 2DOMs might be incorporated into the current technological landscape are cladding around optical fibres, or translating electrical signals into optical pulses. This would be a great step forward, as the expertise needed to design and build these devices would still be required to eventually create all-optical computing chips. Whilst small and fast optical modulators have been produced, with their modulation depth greatly improved by the use of plasmons, many key barriers remain to their commercial use. However, it is hoped that with further exploration of different plasmonic modes and the incorporation of new 2D crystals, these barrier can be overcome.

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